

2-AMALIY ISH

Mavzu: MathCAD integrallash sohasida muhandislik masalalarini yechish.

Ishning maqsadi:

Ushbu amaliy ishning maqsadi talabalarda MathCAD dasturiy muhiti yordamida aniq va taqribiy integrallarni hisoblash bo'yicha amaliy ko'nikmalarni shakllantirishdan iborat. Amaliy mashg'ulot davomida talabalar integrallash tushunchasining muhandislik masalalaridagi ahamiyatini chuqurroq anglaydi, MathCAD muhitida matematik ifodalarni kiritish, hisoblash va natijalarni tahlil qilish imkoniyatlarini o'zlashtiradi. Shuningdek, ushbu ish orqali talabalar real texnik jarayonlarda uchraydigan integral masalalarni kompyuter yordamida samarali yechish malakasiga ega bo'lishi kutiladi.

ADadADASDA

Nazariy qism:

Integrallash matematik analizning eng muhim bo'limlaridan biri bo'lib, u ko'plab muhandislik masalalarining asosini tashkil etadi. Integral yordamida fizik kattaliklarning yig'indisi, yuzalar, hajmlar, ish, energiya, massa va boshqa muhim parametrlar aniqlanadi. Texnik tizimlarda integrallash, ayniqsa, mexanika, elektrotexnika, issiqlik texnikasi va avtomatlashtirish sohalarida keng qo'llaniladi.

The screenshot shows the MathCAD software interface. At the top, there is a toolbar with various icons for solving, text, image, and mathematical operations. Below the toolbar, the main workspace contains a complex formula for $\sigma_q(z)$ involving arctangent and square root functions. Below this formula, there is a text block explaining the variables: q - uniform pressure applied to a rectangular foundation, B - transverse dimension of a rectangular foundation, and L - longitudinal dimension of a rectangular foundation. A note states: "The formula above describes an average value of stresses under a rectangular area." Below this, a double integral formula for $\sigma_v(z)$ is shown, enclosed in a red rectangular box. The formula is:
$$\sigma_v(z) = \int_0^B \int_0^L \frac{3 \cdot w \cdot z^3}{2 \cdot \pi \cdot (x^2 + y^2 + z^2)^{5/2}} dx dy \rightarrow \int_0^B \int_0^L \frac{3 \cdot w \cdot z^3}{2 \cdot \pi \cdot (x^2 + y^2 + z^2)^{5/2}} dx dy$$

Introduction

Compute the test time required to estimate the Bit Error Rate (BER) on a transport to a given confidence level.

Analysis

Definition of the Confidence level approximation

Starting Values

$CL := 0.99$	Confidence Level
$N := 0$	Number of Errors During Test
$\rho := 10^{-10}$	Required BER (Bit Rate Error)
$n := 10^{11}$	Number of Bits Tested

Nonlinear Function Specification

$$n = \frac{-\ln(1-CL)}{\rho} + \frac{\ln\left(\sum_{k=0}^N \frac{(n \cdot \rho)^k}{k!}\right)}{\rho}$$

$$n_{Ref}(CL, N, \rho) := \text{find}(n)$$

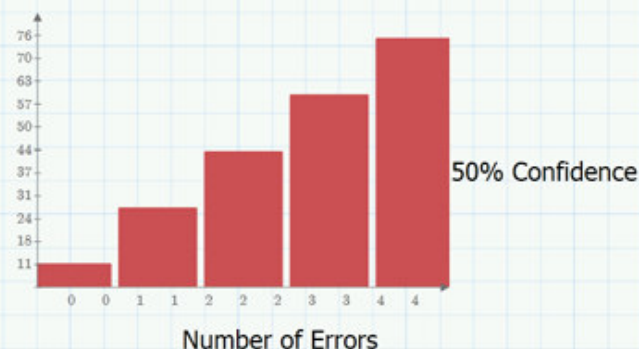
test function based upon solve block

$$T_B(f_{Rate}, CL, N, \rho) := \frac{n_{Ref}(CL, N, \rho)}{f_{Rate}}$$

Example calculation

$$f_{Rate} := 622 \cdot 10^6 \cdot \text{Hz} \quad \rho := 10^{-10}$$

$$T_B(f_{Rate}, .50\%, 1, \rho) = 74 \text{ s}$$



Aniq integral funksiyaning berilgan oraliqdagi umumiy ta'sirini yoki yig'indisini aniqlashga xizmat qiladi. Masalan, tezlik funksiyasi berilgan bo'lsa, uning vaqt bo'yicha integrali yo'lni aniqlash imkonini beradi. Xuddi shuningdek, tok kuchining vaqt bo'yicha integrali elektr zaryadini aniqlashda qo'llaniladi. Shu sababli integral tushunchasi muhandislik hisoblarida muhim o'rin egallaydi.

MathCAD — bu muhandislik hisoblarini bajarish uchun mo'ljallangan kuchli amaliy dastur bo'lib, u matematik formulalarni odatiy yozilish ko'rinishida kiritish imkonini beradi. MathCAD muhitida foydalanuvchi formulani qanday yozsa, dastur ham xuddi shunday ko'rinishda uni ekranga chiqaradi. Bu esa hisob-kitoblarni tushunishni ancha yengillashtiradi.

MathCAD dasturida integrallarni hisoblash uchun maxsus integral operatorlar mavjud. Dastur aniq integralni analitik yoki sonli usulda hisoblab bera oladi. Agar integral analitik yechimga ega bo'lmasa, MathCAD uni taqribiy usullar orqali hisoblaydi. Bu holat murakkab muhandislik masalalarida juda muhim ahamiyatga ega, chunki amaliyotda ko'p hollarda aniq formulaviy yechimni topish qiyin bo'ladi.

Integrallash jarayonida funksiyaning turi, integrallash chegaralari va hisoblash aniqligi muhim hisoblanadi. MathCAD foydalanuvchiga bu parametrlarni oson sozlash imkonini beradi. Natijada hisoblash jarayoni tezlashadi va xatolik ehtimoli kamayadi. Shuningdek, MathCAD hisoblangan integral natijalarini grafik ko'rinishda tasvirlash imkoniyatiga ham ega bo'lib, bu natijalarni tahlil qilishda katta qulaylik yaratadi.

Muhandislik masalalarida integrallash, masalan, detal yuzasini aniqlash, egri chiziq ostidagi maydonni hisoblash, yuklanish ta'sirida bajarilgan ishni topish, issiqlik oqimini aniqlash kabi vazifalarda qo'llaniladi. Shu bois MathCAD integrallash imkoniyatlarini o'zlashtirish bo'lajak mutaxassislar uchun muhim hisoblanadi.

Amaliy ishni bajarishi:

Chiziqli tenglamalar tizimi.

Quyida aniqlangan muammoni o'qing va keyin quyidagi usullar yordamida yechimni ko'rib chiqamiz:

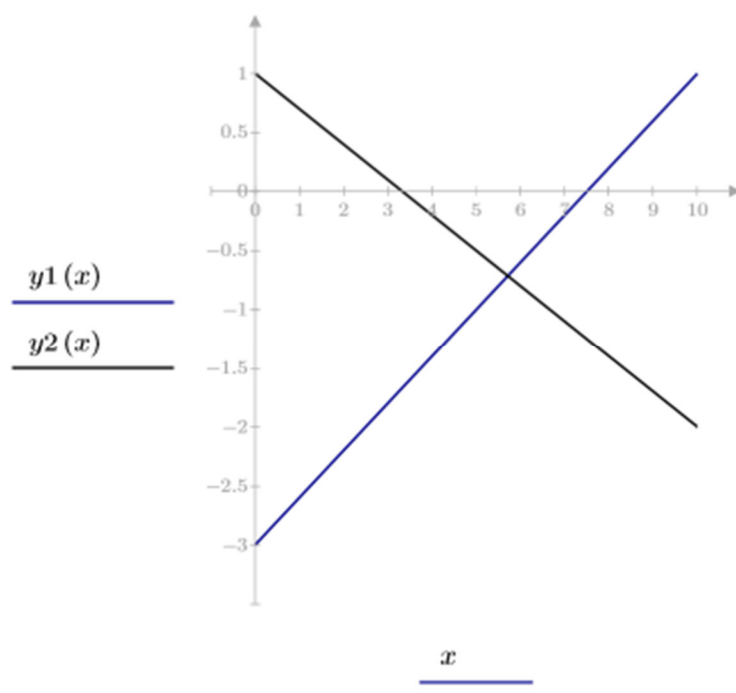
Masalaning qo'yilishi

Quyidagi funksiyalar chiziqli funksiyalardir:

$$y1(x) := \frac{1}{2.5} \cdot (x - 7.5)$$

$$y2(x) := -0.3x + 1$$

Ular quyidagi diagrammada ko'rsatilgandek kesishadi:



Quyidagi tenglamalar to'g'ri bo'lgan kesishish nuqtasining (x, y) koordinatalarini topish kerak:

$$y = \frac{1}{2.5} \cdot (x - 7.5)$$

$$y = -0.3x + 1$$

Tenglamalarni o'zgaruvchilar tenglamalarning chap tomonida bo'lishi uchun qayta tartiblash mumkin:

$$x - 2.5 y = 7.5$$

$$0.3 x + y = 1$$

Tenglamalarni vektorlar va matritsalar orqali qayta yozish mumkin:

$$\begin{bmatrix} 1 & -2.5 \\ 0.3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7.5 \\ 1 \end{bmatrix}$$

Yuqoridagi tenglamadagi har bir vektor va matritsa o'zgaruvchi bilan ifodalanishi mumkin:

$$M \cdot X = v$$

M va v massivlari ma'lum, ammo X noma'lum. X kesishish nuqtasining x va y koordinatalarini ifodalovchi 2 elementli vektordir.

Amaliy ishni bajarilishi:

Matritsa hisoblash bilan yechish.

MathCAD online <https://smath.com/en-US/cloud/sheet/qSZZ9QK8AS> linkidan foydalangan xolda ushbu amaliy ishlarni bajaramiz.

1. M matritsasi va v vektorini aniqlang.

$$M := \begin{bmatrix} 1 & -2.5 \\ 0.3 & 1 \end{bmatrix} \quad v := \begin{bmatrix} 7.5 \\ 1 \end{bmatrix}$$

2. X ni M matritsasi va v vektorining teskari ko'paytmasi sifatida aniqlang.

$$X := M^{-1} \cdot v$$

3. X ni baholang.

$$X = \begin{bmatrix} 5.714 \\ -0.714 \end{bmatrix}$$

Kesish nuqtasining x qiymati 5,714 ga, y qiymati esa -0,714 ga teng.

Yechimchi

Yechimlar ma'lum masalalarni yechish uchun funksiyalardir. Kesish nuqtasining koordinatalarini topish uchun **Solve** o'rnatilgan funksiyasidan foydalanishingiz mumkin.

1. M matritsasini va v vektorini aniqlang.

$$M := \begin{bmatrix} 1 & -2.5 \\ 0.3 & 1 \end{bmatrix} \quad v := \begin{bmatrix} 7.5 \\ 1 \end{bmatrix}$$

2. **lsolve** funksiyasini chaqiring.

$$\text{lsolve}(M, v) = \begin{bmatrix} 5.714 \\ -0.714 \end{bmatrix}$$

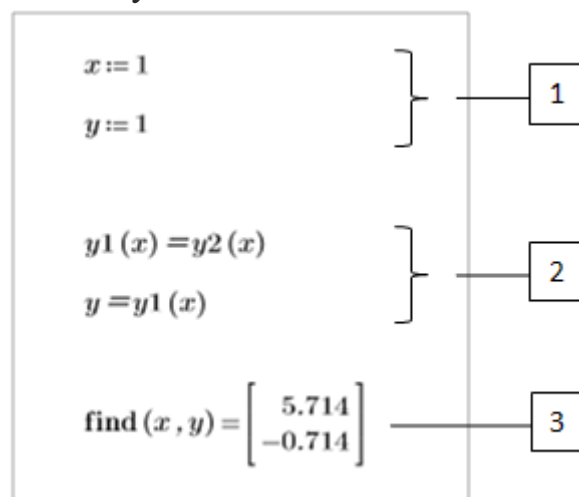
Yechim bloki bilan yechish

Yechim bloki - bu muammoni natural yozuv yordamida aniqlaydigan soha. Matritsa hisoblash va yechuvchi uchun qilganingizdek, tenglamalarni qayta tartiblashingiz shart emas. $y1$ va $y2$ chiziqli funksiyalarini eslang:

$$y1(x) := \frac{1}{2.5} \cdot (x - 7.5)$$

$$y2(x) := -0.3x + 1$$

Quyidagi yechish bloki ikkita funksiyaning kesishish nuqtasini hisoblash uchun **topish** funksiyasidan foydalanadi:



Bu yerda:

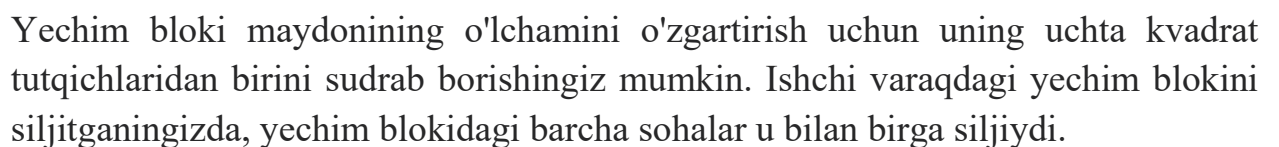
1. Har bir noma'lum uchun qiymatni taxmin qiling
2. Har bir noma'lum uchun cheklov
3. Blok funksiyasini yeching

1. Yechim bloki yordamida kesishish nuqtasining koordinatalarini topish uchun avval ish varag'ingizda $y1$ va $y2$ ikkita funksiyani aniqlashdan boshlang.

$$y1(x) := \frac{1}{2.5} \cdot (x - 7.5)$$

$$y2(x) := -0.3x + 1$$

uchun **Matematika** yorlig'idagi **Mintaqalar** guruhida **Yechim bloki** tugmasini bosing.


$$x := 1$$

4. Muammoni cheklovchi cheklovlarni yozing. Cheklovlarni aniqlaganda Boolean operatorlaridan foydalanishingiz kerak. Boolean teng operatorini kiriting.

$$x := 1$$
$$y := 1$$
$$y1(x) = y2(x)$$
$$y = y1(x)$$

Birinchi cheklov x ning kesishish nuqtasidagi qiymatini, ikkinchi cheklov esa y ning shu nuqtadagi qiymatini belgilaydi.

5. Yechim bloki funksiyasi nomi va argumentlarini kiriting. Bu yerda **find** yozing, keyin funksiya argumentlari sifatida x va y yozing. **find** yorlig'i avtomatik ravishda *kalit so'zga* o'rnatiladi.

$$x := 1$$
$$y := 1$$
$$y1(x) = y2(x)$$
$$y = y1(x)$$
$$\text{find}(x, y)$$

6. Yechim blokini baholang.

$$x := 1$$
$$y := 1$$
$$y1(x) = y2(x)$$
$$y = y1(x)$$
$$\text{find}(x, y) = \begin{bmatrix} 5.714 \\ -0.714 \end{bmatrix}$$

Talabalar uchun individual topshiriqlar:

1 topshiriq:

1. $y = 2x + 5$
2. $y = 3x - 7$
3. $y = 5x$
4. $y = 4x + 9$
5. $y = 7x - 21$
6. $y = 6x + 12$
7. $y = 9x$
8. $y = x - 8$
9. $y = 10x + 20$
10. $y = 2x - 6$
11. $y = x^2 - 9$
12. $y = x^2 + 5x + 6$
13. $y = x^2 - 4x$
14. $y = x^2 - 16$
15. $y = x^2 + 2x - 8$
16. $y = x^2 - 7x + 10$
17. $y = x^2 + x - 12$
18. $y = x^2 - 5x$
19. $y = x^2 + 9$
20. $y = x^2 - 2x - 3$
21. $y = x/2 + 3$
22. $y = 5/x$
23. $y = \sqrt{x} - 4$
24. $y = (2x + 1)/\sqrt{x}$
25. $y = \sqrt{x + 5}$
26. $y = \ln(x)$
27. $y = \ln(x^3) - 27$
28. $y = e^x - 1$
29. $y = x^4 - 16$
30. $y = \log_{10}(x) - 1$

2 topshiriq:

1. $\int_0^\pi (x^2 + 1) dx$ integralini MathCADda hisoblang.

2. $\int_1^3 (2x + 5) dx$ integralining qiymatini toping.
3. $\int_0^\pi \sin x \, dx$ integralini hisoblang va grafigini chizing.
4. $\int_0^1 (3x^2 - 2x + 1) \, dx$ integralini aniqlang.
5. $\int_2^5 x^3 \, dx$ integralini MathCAD'da hisoblang.
6. $\int_0^4 \sqrt{x} \, dx$ integralini toping.
7. $\int_1^e \frac{1}{x} \, dx$ integralini hisoblang.
8. $\int_0^2 (x^2 + 2x) \, dx$ integralini aniqlang.
9. $\int_0^{\pi/2} \cos x \, dx$ integralini hisoblang.
10. $\int_1^2 (5x^2) \, dx$ integralini toping.
11. $\int_0^1 e^{-x^2} \, dx$ integralini MathCAD'da taqribiy usul bilan hisoblang.
12. $\int_0^2 \sin x \, dx$ integralini sonli integrallash orqali toping.
13. $\int_1^3 \ln x \, dx$ integralini taqribiy hisoblang.
14. $\int_0^1 \sqrt{1 - x^2} \, dx$ integralini sonli usulda aniqlang.
15. $\int_0^2 (x^3 + 1) \, dx$ integralini taqribiy hisoblash.
16. $\int_0^5 \frac{1}{1+x^2} \, dx$ integralini hisoblang.
17. $\int_0^3 \sin(2x) \, dx$ integralini taqribiy aniqlang.
18. $\int_1^4 \frac{1}{x^2} \, dx$ integralini sonli integrallash bilan toping.
19. $\int_0^2 e^x \, dx$ integralini MathCAD'da hisoblang.
20. $\int_0^1 \cos(x^2) \, dx$ integralini taqribiy aniqlang.
21. Tezlik funksiyasi $v(t) = 2t + 1$ bo'lsa, $t = 0$ dan $t = 4$ gacha bosib o'tilgan yo'lni toping.
22. Tok kuchi $i(t) = 3t$ bo'lsa, $0 \leq t \leq 2$ oralig'ida o'tgan zaryadni aniqlang.
23. Kuch $F(x) = 5x$ bo'lsa, $x = 0$ dan $x = 3$ gacha bajarilgan ishni hisoblang.
24. Zichlik funksiyasi $\rho(x) = x + 1$ bo'lsa, $0 \leq x \leq 2$ oralig'ida massani aniqlang.
25. Issiqlik oqimi $q(t) = 4t^2$ bo'lsa, $t = 0$ dan $t = 3$ gacha ajralgan issiqlik miqdorini toping.
26. Bosim funksiyasi $p(x) = 2x^2$ bo'lsa, $x = 0$ dan $x = 2$ gacha ta'sir kuchini hisoblang.
27. Tezlanish $a(t) = 6t$ bo'lsa, $t = 0$ dan $t = 2$ gacha tezlik o'zgarishini aniqlang.
28. Tok kuchi $i(t) = \sin t$ bo'lsa, $0 \leq t \leq \pi$ oralig'ida zaryadni toping.

29. Kuch $F(x) = x^2 + 1$ bo'lsa, $x = 1$ dan $x = 3$ gacha bajarilgan ishni aniqlang.
30. Tezlik funksiyasi $v(t) = \cos t$ bo'lsa, $0 \leq t \leq \pi/2$ oralig'ida yo'lni hisoblang.

Manbalar:

1. https://support.ptc.com/help/mathcad/r11.0/en/index.html#page/PTC_Mathcad_Help/about_contour_plots.html#
2. https://support.ptc.com/help/mathcad/r11.0/en/index.html#page/PTC_Mathcad_Help/about_the_mathcad_help_center.html
3. https://support.ptc.com/help/mathcad/r11.0/en/index.html#page/PTC_Mathcad_Help/Tutorials/slv_tutorial/task1-1_linear_system_of_equations.html# misol

Nazorat savollari:

1. Integral tushunchasi nimani ifodalaydi?
2. Aniq integralning muhandislikdagi ahamiyati nimada?
3. MathCAD dasturi qanday vazifalarni bajarishga mo'ljallangan?
4. MathCAD muhitida integral qanday kiritiladi?
5. Taqribiy integrallash nima sababdan qo'llaniladi?
6. Integral yordamida qaysi fizik kattaliklar aniqlanishi mumkin?
7. Grafik tasvir integral natijasini qanday tushunishga yordam beradi?
8. MathCAD dasturining muhandislik hisoblaridagi afzalliklari nimalardan iborat?
9. Integrallash jarayonida aniqlik nima uchun muhim?
10. MathCAD yordamida olingan natijalarni tahlil qilish qanday amalga oshiriladi?